BUILD GUIDE · V1.0.1

Desk Buddy

A small desk companion: a face that blinks and loves to be petted, a focus timer, clock & weather, and a Claude Code monitor.



Living face

Blinks, glances around, gets happy when you turn the knob or pet the pad.

Focus timer

25/5 minute Pomodoro; the face turns serious while you focus.

Clock & weather

Synced over the internet, with weather for your location.

Claude Code monitor

Context, session & weekly limits, alarm when Claude needs permission.

Firmware & code: https://github.com/julisman/desk-buddy · Installer: https://julisman.github.io/desk-buddy/

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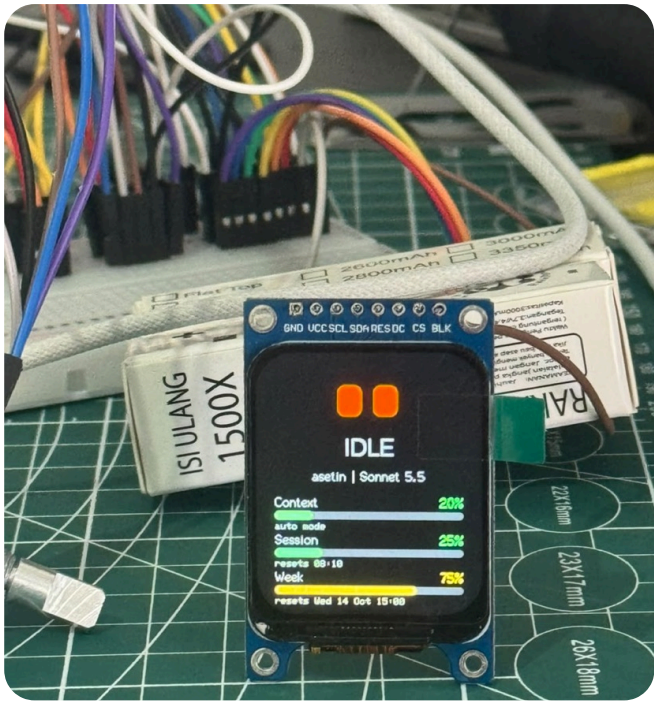
Meet Desk Buddy

A 1.69" screen with a lively robot face. Hold the knob to switch modes.



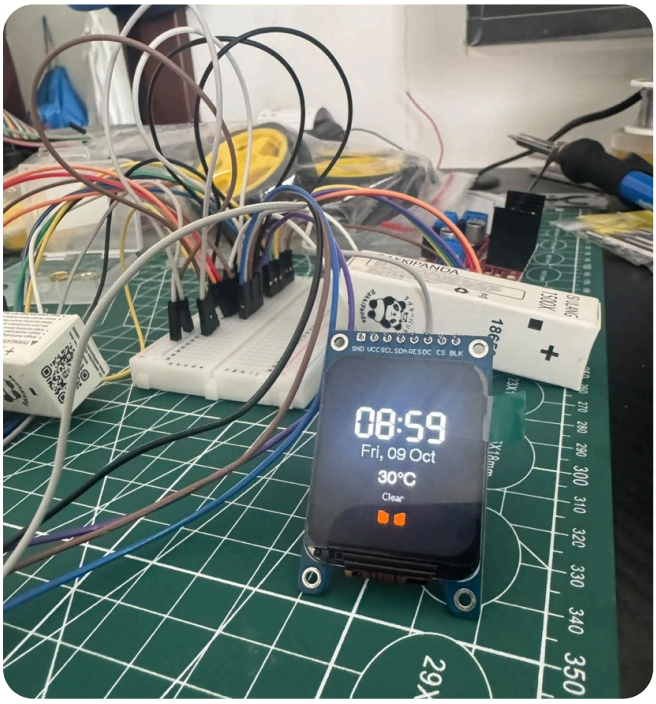
CLAUDE · WORKING

Claude Code is working: context, session and weekly limits, reset times.



CLAUDE · IDLE

The session is waiting for the next prompt. Bar colours: green is safe, yellow is getting high, red is almost used up.



CLOCK

Time, date and weather for your location.



FOCUS

Pomodoro with a progress ring; the face turns serious.

Parts list

Approximate Shopee prices (October 2026). Search for the keywords directly on Shopee.

PART	KEYWORD	PRICE	NOTES
ESP32-S3 N16R8 DevKitC (2 USB-C ports)	`esp32 s3 n16r8`	Rp85–110k	Must be N16R8 (16MB flash, 8MB PSRAM)
LCD 1.69" ST7789 240x280 SPI	`lcd 1.69 st7789`	Rp40–55k	One with a BLK pin
Rotary encoder KY-040	`ky-040 rotary encoder`	Rp6–10k	Turn + press knob
Passive buzzer KY-006	`ky-006 buzzer pasif`	Rp4–7k	3-pin module: S, +, –
Breadboard, 400 holes	`breadboard 400`	Rp8–12k	Or a mini breadboard
Female-male jumper wires (20 pcs)	`kabel jumper female male`	Rp8–12k	Signals to the ESP32 pins
Male-male jumper wires (20 pcs)	`kabel jumper male male`	Rp8–12k	To the breadboard power rails
USB-C data cable	`kabel usb c data`	Rp15–25k	Not a charge-only cable
Copper tape (optional)	`copper tape`	Rp10–15k	Touch pad for petting
Estimated total			**Rp185–260k**

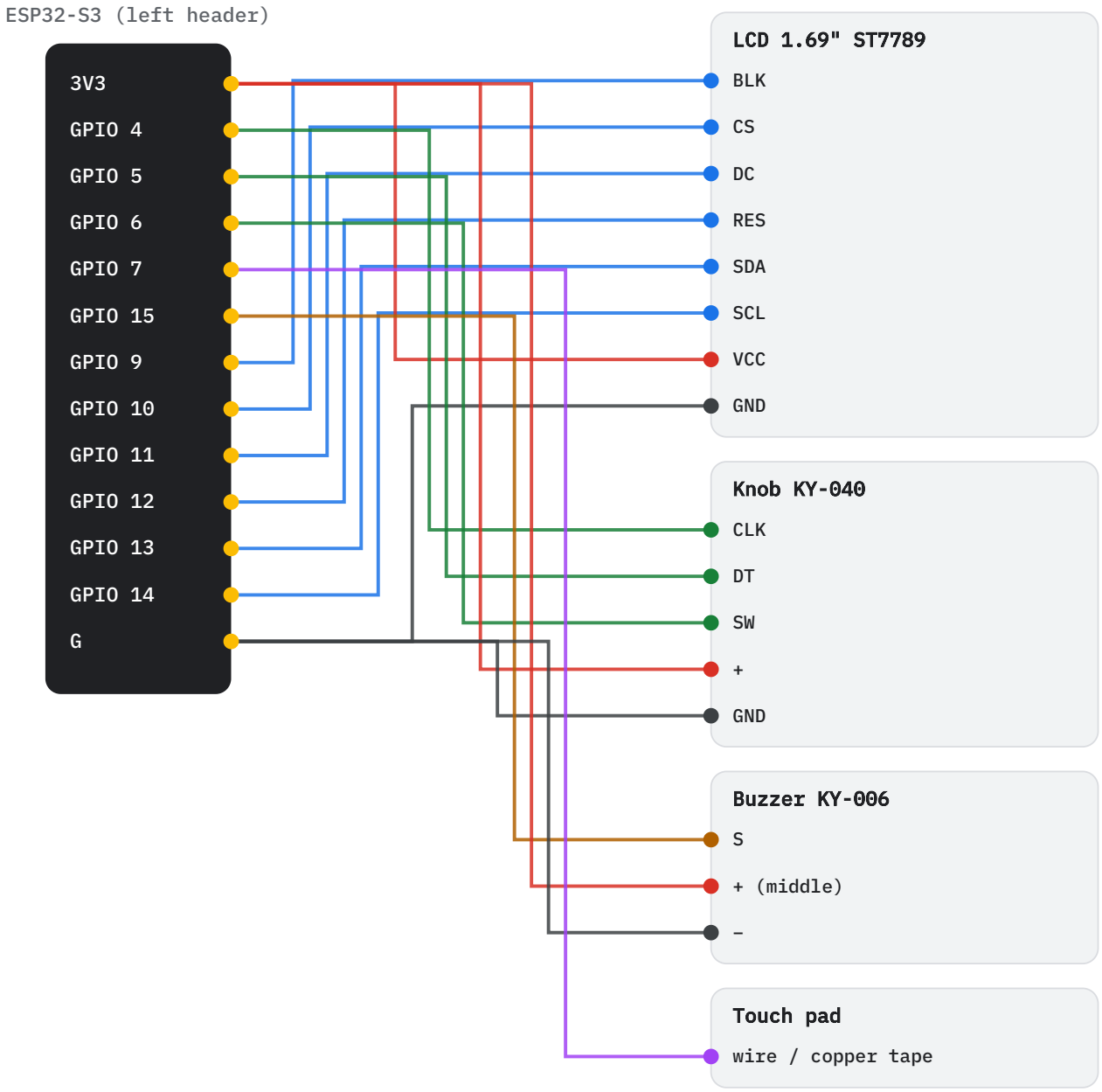
Tools

A laptop/PC with Google Chrome or Microsoft Edge (to install the firmware). No soldering needed for this breadboard version. Optional: solder the LCD header so the contacts last.

> **Important:** the board must be an **ESP32-S3 N16R8**. Other variants (N8, N8R2, plain ESP32) have different memory and this firmware will not run on them.

Wiring

All signals go to the left header of the ESP32-S3. Power is shared through the breadboard rails: pin **3V3** to the **+** rail, pin **G** to the **–** rail.



Connection table

MODULE	MODULE PIN	CONNECT TO
LCD 1.69" ST7789	BLK	GPIO 9
LCD 1.69" ST7789	CS	GPIO 10
LCD 1.69" ST7789	DC	GPIO 11
LCD 1.69" ST7789	RES	GPIO 12
LCD 1.69" ST7789	SDA	GPIO 13
LCD 1.69" ST7789	SCL	GPIO 14
LCD 1.69" ST7789	VCC	3V3 (breadboard + rail)
LCD 1.69" ST7789	GND	GND (breadboard – rail)
Knob KY-040	CLK	GPIO 4
Knob KY-040	DT	GPIO 5
Knob KY-040	SW	GPIO 6
Knob KY-040	+	3V3 (breadboard + rail)
Knob KY-040	GND	GND (breadboard – rail)
Buzzer KY-006	S	GPIO 15
Buzzer KY-006	+ (middle)	3V3 (breadboard + rail)
Buzzer KY-006	–	GND (breadboard – rail)
Touch pad	wire / copper tape	GPIO 7

> **Do not use the 5V pin.** When plugged into the native USB port, the 5V pin has no power. All modules use 3V3.

Assembly steps

1. Unplug USB. Push the LCD, knob and buzzer into the breadboard, each in its own rows.
2. Male-male wires: ESP32 pin **3V3** to the **+** rail, pin **G** to the **–** rail. Then each module's VCC/+ to the + rail and GND/– to the – rail.
3. Female-male wires: female end on the ESP32 pin, male end in the module's pin row, as in the table.
4. Touch pad: one wire to GPIO 7. Leave the other end free, or stick on 2x2 cm of copper tape (it works through a plastic case).
5. Double-check: no wire on the 5V pin, and the + and – rails are not swapped.

> LCD without soldering: the LCD header pins often make poor contact. Insert it slightly tilted and hold it with tape so the pins press against the sides of the holes.

Install the firmware

No software to install. On your laptop, open (Chrome or Edge):

`https://julisman.github.io/desk-buddy/`

1. Plug the ESP32-S3 into your laptop with a USB-C **data** cable, using the port labelled **USB** (not COM).
2. Click **Install Desk Buddy**, choose the port that appears (usually "USB JTAG/serial debug unit"), then **Install**. Wait about 1 minute.
3. Press the **RST** button on the board. The screen flashes red-green-blue, then the face appears.

Connect to WiFi

1. The bottom of the screen shows the setup banner `WiFi setup: join DeskBuddy-XXXX` / `then open 192.168.4.1`, where `XXXX` is your device's code, e.g. `DeskBuddy-1A2B`.
2. On your phone, open the WiFi settings and join that hotspot. The setup page opens by itself; if it doesn't, open `192.168.4.1` in the browser.
3. Fill in `WiFi name` (your home WiFi), `Password`, `Time zone` (e.g. WIB/WITA/WIT) and `Weather location` (your city), then tap `Save & connect`. The page shows `Saved`, Desk Buddy restarts and connects.

> Changing WiFi / moving house: hold the knob while powering on Desk Buddy until `WiFi reset` appears. If the saved WiFi fails to connect, the setup hotspot also opens by itself.

Want to change the code? The full source is at `https://github.com/julisman/desk-buddy` (PlatformIO).

How to use

Hold the knob (about 1 second) to change mode. **Pet** the touch pad any time: the face gets happy, hearts pop out, and it "purrs".

FACE
Turn = petting (happy, glancing). Press = boop (surprised).

FOCUS
Turn = set minutes. Press = start / pause. Focus for 25 minutes, then a 5-minute break starts automatically.

CLOCK
Time, date and weather for your location. Before WiFi is set up it shows `WiFi setup (see below)`.

CLAUDE
Claude Code status from your laptop. Turn = switch session. Press = silence the alarm.

Untouched for 5 minutes (at night, 23:00–07:00: 1 minute), Desk Buddy goes to sleep and dims the screen. Touch the knob to wake it.

CLAUDE mode (for Claude Code users)

Desk Buddy shows the model, context %, session (5-hour) and weekly limits with their reset times, and the status: **WORKING**, **NEEDS YOU** (alarm + red frame when Claude asks for permission), **DONE**. It can track up to 4 sessions at once.

1. Laptop and Desk Buddy on the same WiFi. Requires `jq` (`brew install jq`).
2. Download the source code from https://github.com/julisman/desk-buddy, then run `tools/install-claude-hooks.sh`. To remove it: `--uninstall`.
3. Start a new Claude Code session. To try it without Claude: `tools/claude-desk-buddy.sh demo`.

Only numbers and status are sent to Desk Buddy. Your prompts and Claude's replies never leave your laptop.

Troubleshooting

SYMPTOM	FIX
Screen lights up but is plain grey / white	The LCD data wires (SDA, SCL, CS, DC, RES) aren't making contact. Re-seat them, tilt the LCD header slightly in the breadboard, hold it with tape, press RST. Soldering the LCD header is the permanent fix.
Screen completely dead	LCD VCC must go to 3V3, not 5V (the 5V pin has no power when plugged into native USB).
Board keeps restarting when it beeps	Check the buzzer wiring: S to GPIO 15, + to 3V3, – to GND. The firmware already uses a low volume to be safe.
Port doesn't appear when installing	Swap the USB-C cable (many are charge-only). Try the other USB port. Or hold BOOT, press RST, release BOOT.
Knob reversed / jumps by 2	Swap the CLK and DT wires. If 1 click = 2 steps, let us know (firmware setting).
No clock / weather	WiFi isn't connected yet (clock mode shows `WiFi setup (see below)`). Hold the knob while powering on to redo WiFi setup.
Petting isn't detected	Don't touch the pad while Desk Buddy is powering on (calibration). Use a short wire and a wider pad (2x2 cm copper tape).

Next

Phase 2: LiPo battery + Type-C charger (take it anywhere, battery icon) and voice: talk to Desk Buddy through a mic and speaker, answered by an AI running on your laptop.